

Two simple lessons of unitarity: on the equivalence principle in black hole physics and the Born rule

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Abstract

In this paper, we demonstrate surprisingly simple yet powerful lessons provided by unitarity in quantum physics as applied to effective theory. While fine-grained quantum gravity states can be significantly different from coarse-grained effective-theory states, if unitarity is respected on the effective side, then the no-drama equivalence principle is incompatible with the black hole exterior's eventual purification. This holds regardless of whether the effective-side unitary evolution is state-independent or state-dependent. One of the three has to be given up - equivalence principle, effective bulk Hilbert space factorization, or, most importantly and previously neglected, unitarity of an effective theory. Additionally, the Born rule can be derived under maximally chaotic interactions, as long as a coarse-grained observable selects the pointer basis.

1 Introduction

What drives the results in this paper are two sets of equations, both underpinned by unitarity in quantum physics and set entirely within the effective theory's Hilbert space, though some of the states involved are more coarse-grained than others. Despite having consequences for gravity, and being inspired by quantum gravity, none of the mathematical details require quantum gravity; ordinary unitary quantum mechanics suffices.

(1) Effective-side unitarity + no-drama equivalence principle validates the semi-classical Hawking results for black hole evaporation.

All states $|\psi\rangle \in \mathcal{H}_{eff}$ ($|\psi\rangle, |\phi\rangle$, etc.),

t refers to exterior observer time,

Δt being evolution time interval used in unitary $U_{eff,t}, U_{EFT}$,

T being the current exterior observer time,

$S(t)$ roughly being coarse-grained entropy, large at every t ,

$$S_{\min} \equiv \min_{0 \leq t \leq T} S(t), \tag{1}$$

$$|\alpha_{\psi_{EFT,t}}|^2 = 1 - e^{-S(t)} \approx 1,$$

$$\langle \psi_{EFT,t} | \phi_{correction,t} \rangle = 0.$$

$$U_{eff,t} |\psi_{EFT,t}\rangle = \alpha_{\psi_{EFT,t}} U_{EFT} |\psi_{EFT,t}\rangle + \sqrt{1 - |\alpha_{\psi_{EFT,t}}|^2} |\phi_{correction,t}\rangle \Rightarrow$$

$$U_{eff,t} |\psi_{EFT,t}\rangle \approx U_{EFT} |\psi_{EFT,t}\rangle \Rightarrow$$

$$U_{eff,T} U_{eff,T-\Delta t} \dots U_{eff,0} |\psi_{EFT,0}\rangle \approx (U_{EFT})^{T+1} |\psi_{EFT,0}\rangle \quad \text{for } T \ll e^{S_{\min}/2} \text{ (by unitarity)}$$

(2) The Born rule can be derived from maximally chaotic interactions. (chaotic justification of the envariance argument)

H_I interaction Hamiltonian, $\hat{O}_S$ coarse-grained system measurement observable,

$\hat{P}_S$ is the projection operator for $\hat{O}_S$,

H_I being maximally chaotic interaction,

$|\Psi_i\rangle$ is the initial pre-interaction state of the universe,

$|\Psi_f\rangle$ is the final post-interaction state of the universe,

$|\psi_{S,k}\rangle$ is a fine-grained pointer basis for system S ,

$N_E \approx \infty$ is the number of environment degrees of freedom, (2)

$$H = H_S + H_E + H_I,$$

$$[\hat{O}_S, H_I] = 0,$$

$$\langle \Psi_i | \hat{P}_S(\lambda_j) | \Psi_i \rangle = p_j \Rightarrow$$

$$|\Psi_f\rangle \approx \frac{1}{\sqrt{N_E}} \sum_{k=1}^{N_E} e^{i\varphi_k} |\psi_{S,k}\rangle |\phi_{E,k}\rangle \Rightarrow$$

$$[\text{The number of } \psi_{S,k} \text{ compatible with } \hat{P}_S(\lambda_j)] \approx N_E p_j \text{ (by unitarity)} \Rightarrow$$

Born rule, assuming equal probability for $|\psi_{S,k}\rangle$ in $|\Psi_f\rangle$.

The rest of this paper fills in the derivation details and their interpretation; both revolve around the consequences of **unitarity of an effective quantum theory**, which is almost always implicitly assumed and yet is a stronger condition than unitarity of the full quantum gravity theory.

Some consequences and caveats are briefly noted below.

For Equation (1), A) $U_{eff,t}$ is allowed to be state-dependent and time-dependent from the start, so the result cannot be avoided by resorting to state-dependent maps, including state-dependent error-correcting codes. B) $U_{eff,t}$ works solely on $\mathcal{H}_{eff}$, the effective Hilbert space, so the objection that ‘quantum gravity states are structurally different from effective states’ (invoking the eigenstate thermalization hypothesis: ETH) cannot be used to avoid the result. Since we allow for state-dependent and time-dependent corrections by the full quantum gravity theory, non-existent entanglement in the full quantum gravity theory cannot suddenly appear as significant entanglement in the effective theory; indeed, this is why the ‘information problem’

is named for information in the first place. C) The result of Equation (1) suggests that we have to give up at least one of the three - **equivalence principle, (bulk) Hilbert space factorization, or, most importantly and often neglected, unitarity of an effective theory.**

For Equation (2), the Born rule derivation assumes that a measurement process involves maximally chaotic interactions, which may not be justified. Furthermore, the measurement problem is not really resolved, and while we can resort to quantum gravity intuitions for why the pointer basis becomes the measurement basis, this remains incomplete for now.

2 (1) On effective-side unitarity + no-drama equivalence principle validating the semiclassical Hawking results

2.1 Interpreting Equation (1)

Translated to words from Equation (1): **even if** the unitary effective theory is **state-dependent and time-dependent** around the black hole horizon, especially at late-time black hole evaporation, as long as the Δt -corrections around the conventional semiclassical EFT-like state remain exponentially small at every time within the effective theory (no-drama **equivalence principle** / small corrections), the effective-side deviations of the black hole from the original effective field theory are guaranteed to remain **exponentially small** throughout the entire black hole evaporation.

2.1.1 Computational complexity or error correction codes do not help.

This is not about an epistemic issue - we are not dealing with the case where a pure effective-theory state or a pure quantum gravity state cannot be distinguished by an observer from a heavily entangled state. Rather, this is the case where a pure quantum gravity state with the black hole interior fully evaporated away is coarse-grained to a heavily entangled effective-theory state. To maintain the final purity of the evaporation process under the equivalence principle, the natural interpretation can only be that **the bulk Hilbert space factorization breaks down** at the effective theory level. It is not enough to say that the effective bulk encodes the full theory states redundantly; there has to be factorization breakdown - see Section 2.1.3 for why no new entanglement should be created in the corrected effective theory per information conservation reasons.

Somewhat alternatively, this heavily entangled state may be considered to be an ensemble state, instead of an actual effective state [1, 2]. In this view, the effective theory can only maintain the equivalence principle by being an ensemble theory of many possible actual theories, each probabilistically weighted. However, **this view sacrifices the late-time equivalence principle for the actual non-ensemble effective theory** [3].

2.1.2 The structure of quantum gravity states differing from that of effective states does not matter - eigenstate thermalization hypothesis (ETH) is irrelevant.

Worth re-emphasizing is that the states $(|\psi\rangle, |\phi\rangle)$ appearing above are all **in the effective theory's Hilbert space** $\mathcal{H}_{eff}$, and we are not particularly constraining the form of effective states.

Quantum gravity's role in Equation (1) can only be about quantum gravity states being mapped to effective states after whatever corrections are to be made. Therefore, the fact that states of the full quantum gravity theory are expected to differ significantly from effective-theory states is irrelevant to the results.

2.1.3 Proper coarse-graining cannot create new entanglement and relations between subsystems that are non-existent in the quantum gravity theory.

There have been misunderstandings about applications of the eigenstate thermalization hypothesis (ETH) within a quantum gravity theory. ETH is about coarse-grained and simple observables, and it gives an illusion that a pure quantum gravity state is dual to a heavily mixed effective state. But this only says that a pure state and some mixed state may be practically indistinguishable using coarse observables, whether one is working in the effective theory or in the full quantum gravity theory; this is an epistemic issue, not evidence that an effective state is actually heavily mixed.

A quantum error-correcting code analogy can be used here. We may use these codes to create entangled physical states such that the information of pure logical states can be protected from noise. This does not mean that a physical theory is a coarse-grained version of a logical theory; the opposite is true, if anything. Coarse-graining makes existing entanglement invisible to observers but cannot create new entanglement or relations between subsystems. This is an information conservation constraint.

It may be argued that because operators of an effective theory are too simple and limited,

the theory may falsely display new relations. Equation (1) anticipates such an objection, which is why state-dependence and time-dependence are allowed, so that the full quantum gravity theory can always correct the effective theory whenever needed.

The standard ETH ansatz makes the epistemic character of this indistinguishability precise. For a chaotic system with energy eigenstates $|E_i\rangle$ and a simple, few-body operator $\hat{O}$,

$$\langle E_i | \hat{O} | E_j \rangle = O(\bar{E}) \delta_{ij} + e^{-S(\bar{E})/2} f_O(\bar{E}, \omega) R_{ij}, \quad (3)$$

where $\bar{E} = (E_i + E_j)/2$, $\omega = E_i - E_j$, $O(\bar{E})$ is the smooth microcanonical average, f_O is a smooth function, and R_{ij} is an $O(1)$ pseudo-random variable. The diagonal part alone, $\langle E_i | \hat{O} | E_i \rangle \approx O(\bar{E})$, already says that a single, exactly pure energy eigenstate reproduces the thermal expectation value of $\hat{O}$ to within corrections exponentially small in S . Nothing about $|E_i\rangle\langle E_i|$ has become mixed; the state remains exactly pure, and what ETH describes is a property of the matrix elements of $\hat{O}$ evaluated in that pure state, not a property of the state's purity itself. Resolving the difference between $|E_i\rangle$ and a genuinely mixed thermal density matrix requires accessing the off-diagonal structure carried by R_{ij} , suppressed by $e^{-S/2}$ and hence invisible to any operator built from simple, coarse observables. This is exactly the sense in which ETH supports an epistemic reading rather than an ontological one: it explains why coarse measurements cannot tell a pure eigenstate from a thermal ensemble, not why the eigenstate is secretly thermal.

2.1.4 We need to give up one of the three to resolve the information problem - late-time equivalence principle, effective bulk Hilbert space factorization, unitarity of the effective theory.

Exponentially small corrections cannot produce an $O(1)$ difference in effective-theory entanglement entropy - this is also confirmed by the small corrections theorem [4].

2.2 Hilbert space factorization breakdown

The effective Hilbert space factorization breakdown view denies that the effective state $|\psi_{eff}\rangle$ can be fully factorized into black hole exterior and interior components for late-time evaporation.

The islands proposal, when naively interpreted, gives up the effective bulk Hilbert space factorization, since, for late-time evaporation, the entanglement wedge of the exterior radiation

comes to include the black hole interior islands as well.

The proposal (which is upheld by the replica wormhole approach as well) computes the generalized entropy of the radiation R as

$$S(R) = \min_X \text{ext}_X \left[\frac{\text{Area}(X)}{4G_N} + S_{\text{semicl}}(R \cup I_X) \right], \quad (4)$$

extremizing and then minimizing over candidate quantum extremal surfaces X homologous to $R \cup I_X$, so that R and X together bound a bulk region containing the island I_X . Before the Page time, the trivial (empty) island dominates and reproduces the ever-increasing semiclassical answer; after the Page time, a nontrivial island, comprising a portion of the black hole interior behind the horizon, dominates instead, and $S(R)$ turns over and saturates at the Page curve's plateau. It is the appearance of this nontrivial island, extending the entanglement wedge of R into what was previously the interior, that constitutes the naive Hilbert space factorization breakdown referred to above.

An equivalent formulation dispenses with the separately added S_{semicl} term altogether. Extremizing the area functional directly on the appropriately continued gravitational replica geometry, including its backreaction from bulk matter, reproduces the same generalized entropy as the area of a single quantum-corrected extremal surface [5, 6].

The problem, however, has been that the factorization breakdown interpretation has not been settled. In fact, if we take the breakdown interpretation seriously, this suggests that quantum gravity naturally does not admit bulk Hilbert space factorization, and that this becomes visible even at the effective level during late-time black hole evaporation - something resembling holography of information [7]. (Note that this reading of factorization breakdown, arising specifically from the gravitational path integral's inclusion of spacetime wormholes and a resulting baby-universe Hilbert space, has been argued for - see [8] as an example.) But the Page curve [9] is often computed and reproduced within the islands proposal [10, 11], and the Hilbert space factorization breakdown makes entanglement entropy either trivial or ill-defined.

This tension may be avoided within holography by assuming that the boundary theory admits the traditional factorization, thereby rescuing the relevance of the Page curve. However, unless holography holds for the actual universe, this cannot be the full answer for rescuing the Page curve.

2.3 (Late-time) equivalence principle breakdown

An alternative, less mainstream approach has been to deny the equivalence principle. In other words, at late times, we have, instead of Equation (1):

$$\begin{aligned} U_{eff,t}|\psi_{EFT,t}\rangle &\not\approx U_{EFT}|\psi_{EFT,t}\rangle, \quad t, T \text{ being late-time (evaporation)} \Rightarrow \\ U_{eff,T}U_{eff,T-\Delta t}..U_{eff,0}|\psi_{EFT,0}\rangle &\not\approx (U_{EFT})^{T+1}|\psi_{EFT,0}\rangle \end{aligned} \quad (5)$$

This has indeed been the view of fuzzball and firewall proponents [12]. Firewall arguments have become less common and arise more as purported consequences of replica wormholes [13, 14, 15]. Recent work has in fact revived firewall-like conclusions directly from replica-wormhole calculations themselves, rather than from the original complementarity-based argument: old black holes have been argued to tunnel into a white-hole/firewall configuration via large baby-universe emission, with a probability computable within the wormhole expansion [16].

2.4 Non-unitarity of effective theory

Non-unitarity of the effective theory, or non-existence of an effective quantum theory, can be a reasonable position as long as it is properly acknowledged - though this is rarely done. Our standard uses of quantum mechanics all assume unitarity even on the effective-theory side. Quantum computers operate at the effective-theory level, not the quantum-gravity level, and their governing theories all assume unitary evolution at that effective level.

The way non-linear classical theories arise from linear unitary quantum mechanics is not fully understood - chaos in particular - but unitarity of quantum mechanics in the effective theory, at a level finer than the usual scope of classical mechanics, has remained largely unquestioned as an important constraint. Whether classical chaos genuinely emerges from unitary quantum dynamics at all, rather than being an artifact of the classical limit taken too casually, has itself been argued to be a nontrivial and unresolved question, on the grounds that finite quantum systems with discrete spectra cannot sustain the same long-time exponential sensitivity their classical counterparts display [17].

Linearity is not an incidental feature of quantum mechanics that effective theories are free to relax. Generic non-linear modifications to the Schrödinger equation have been shown to permit superluminal, EPR-type signaling between separated systems [18, 19] - an information-theoretic constraint that applies with equal force to an effective theory as to a fundamental one, since the

signaling argument only requires linear evolution to fail somewhere accessible to experiment.

However, it could be that whenever quantum gravity effects are large enough, effective theory can no longer be maintained if we are to maintain the equivalence principle. This does not imply that quantum mechanics is fundamentally broken or incomplete - rather, it only means that a use of quantum mechanics with heavily coarse-grained effective states may show illusions of non-unitarity and superluminal signaling. This, in turn, may have the potential to explain how effective non-linearity emerges from linear quantum mechanics.

2.5 Escape route: a superposition of smooth horizons

Despite the stark trichotomy: give up the equivalence principle, give up factorization, or give up effective unitarity, a more refined modification of the equivalence principle route is available.

Refine the coarse EFT-like state into its constituent microstates:

$$|\psi_{EFT,t}\rangle = \sum_k \alpha_k |\psi_{k,EFT,t}\rangle \quad (6)$$

Equation (1) need not hold uniformly across k . For most k ,

$$U_{eff,t}|\psi_{k,EFT,t}\rangle \approx U_{EFT}|\psi_{k,EFT,t}\rangle \text{ for most } k \quad (7)$$

exactly as before. But a subset $\mathcal{K}_t$ may instead undergo a complete transition,

$$U_{eff,t}|\psi_{k,EFT,t}\rangle = |\psi'_{k,t}\rangle, \quad \langle\psi_{k,EFT,t}|\psi'_{k,t}\rangle \approx 0, \quad k \in \mathcal{K}_t, \quad (8)$$

with $|\psi'_{k,t}\rangle$ an ordinary, horizon-free configuration - an evaporated exterior with no interior left to speak of.

Each individual classical background branch can maintain the equivalence principle.

While individual branches in $\mathcal{K}_t$ have:

$$U_{eff,t}|\psi'_{k,t}\rangle \not\approx U_{EFT}|\psi'_{k,t}\rangle, \quad k \in \mathcal{K}_t, \quad (9)$$

the equivalence principle can be kept in the sense that:

$$U_{eff,t}|\psi'_{k,t}\rangle = U_{no\ BH\ EFT}|\psi'_{k,t}\rangle, \quad k \in \mathcal{K}_t. \quad (10)$$

That is, we distinguish the EFT under stable black holes from the EFT without any black hole (and thus without any horizon to speak of). Each classical background branch then maintains the equivalence principle.

As evaporation proceeds, $\sum_{k \in \mathcal{K}_t} |\alpha_k|^2$ grows monotonically toward $O(1)$ and then toward 1, and the black hole branch is correspondingly driven toward vanishing weight, consistent with a unitarily purified exterior.

This is close in spirit to the fuzzball proposal [20, 21, 22], applied here microstate by microstate with an explicit account of why the coarse aggregate is nonetheless well-approximated by Equation (1) for as long as $\mathcal{K}_t$ remains a small fraction of the ensemble.

2.5.1 The conventional quantum gravity argument naturally leads to a superposition of smooth horizons

Hawking emission is thermal: the energy carried by each quantum is not fixed, but drawn from a distribution. Energy conservation ties the black hole’s mass after k emissions to the specific sequence of energies radiated,

$$M_k = M_0 - \sum_{i=1}^k \omega_i. \quad (11)$$

If gravity responds linearly to the actual, operator-valued matter energy-momentum rather than to its expectation value, a superposition over emission histories $\{\omega_1, \dots, \omega_k\}$ sources a superposition over geometries, each branch carrying its own M_k entangled with the radiation history that produced it.

This observation was not historically taken as the starting point for black hole evaporation. Standard practice instead evolves a single mean-field trajectory, $\bar{M}(t) = M_0 - \int \langle dE/dt \rangle dt$, sourced by $\langle T_{\mu\nu} \rangle$. This is an excellent approximation for coarse observables because such observables are sums of many weakly correlated microscopic contributions, and the true M_k concentrates tightly around $\bar{M}(t)$ by the same mechanism behind any law-of-large-numbers-type result. Combined with the traditional intuition that the true gravity state cannot be so different from the effective state under the equivalence principle, it used to be thought that a superposition of horizons cannot be used to resolve the information problem [23].

Holographic gravity since then has become the dominant mainstream approach to the black hole information problem, and there is largely no issue with a single smooth horizon in holographic gravity. Therefore, the superposition-of-horizons idea was never revisited, even though

the fact that coarse observables cannot distinguish a superposition from a pure state had by then become canonical in quantum gravity, and even though what the idea actually requires is only a subtle, faithful modification of the equivalence principle rather than its complete abandonment.

Once the subtle modification to the equivalence principle is recognized, the idea that a coarse observable cannot distinguish a superposition of horizons from a single horizon becomes a virtue: it suggests why the equivalence principle practically works for the superposition as well.

One objection can be expected in advance: while the ‘superposition of horizons’ is a reasonable (and possibly expected) conceptual point of view when gravity is assumed to be quantized and the equivalence principle is considered largely touchable, this has not been shown in some quantum gravity theory. For now, theoretical demonstrations within an actual quantum gravity theory are left to future work.

2.5.2 The central limit theorem argument for coarse indistinguishability

Let $\hat{E}_{rad}(T) = \sum_{i=1}^{N(T)} \omega_i$ be the total radiated energy up to observer time T , built from $N(T) \gg 1$ semi-independent Hawking quanta, each with bounded per-quantum variance σ^2 . By the Lindeberg-Feller central limit theorem,

$$\frac{\hat{E}_{rad}(T) - \langle \hat{E}_{rad}(T) \rangle}{\sqrt{N(T)} \sigma} \xrightarrow{d} \mathcal{N}(0, 1), \quad (12)$$

so the fractional fluctuation of $\hat{E}_{rad}(T)$ scales as

$$\frac{\text{Var}[\hat{E}_{rad}(T)]}{\langle \hat{E}_{rad}(T) \rangle^2} \sim \frac{1}{N(T)} \rightarrow 0. \quad (13)$$

Any coarse observable built this way, as a sum over $N(T)$ semi-independent contributions, is correspondingly insensitive to the underlying branch structure: the microstates in $\mathcal{K}_t$ contribute to the variance of $\hat{E}_{rad}(T)$ minimally, right because they cannot contribute to radiations by definition. Furthermore, observers cannot practically distinguish a superposition of horizons and horizon-less microstates from single-horizon black hole spacetime.

2.6 Remaining derivations of Equation (1)

We now derive the last equation of Equation (1)

$$U_{eff,T}U_{eff,T-\Delta t}\dots U_{eff,0}|\psi_{EFT,0}\rangle \approx (U_{EFT})^{T+1}|\psi_{EFT,0}\rangle \quad (14)$$

from

$$U_{eff,t}|\psi_{EFT,t}\rangle = \alpha_{\psi_{EFT,t}} U_{EFT}|\psi_{EFT,t}\rangle + \sqrt{1 - |\alpha_{\psi_{EFT,t}}|^2} |\phi_{correction,t}\rangle \quad (15)$$

Take U_{EFT} to propagate the EFT-like state family forward, $U_{EFT}|\psi_{EFT,t}\rangle = |\psi_{EFT,t+\Delta t}\rangle$, and abbreviate $\alpha_t \equiv \alpha_{\psi_{EFT,t}}$, $S_k \equiv S(k\Delta t)$, $|\Psi_k\rangle \equiv U_{eff,k\Delta t} \dots U_{eff,0}|\psi_{EFT,0}\rangle$, $|\Phi_k\rangle \equiv (U_{EFT})^{k+1}|\psi_{EFT,0}\rangle = |\psi_{EFT,(k+1)\Delta t}\rangle$, and $\epsilon_k \equiv |||\Psi_k\rangle - |\Phi_k\rangle||$. $S(t)$ need not be constant in time; only $S(t)$ large at every $t \in [0, T]$ is used below.

Write $|\Psi_{k-1}\rangle = |\Phi_{k-1}\rangle + |\delta_{k-1}\rangle$ with $|||\delta_{k-1}\rangle|| = \epsilon_{k-1}$. Since $|\Phi_{k-1}\rangle = |\psi_{EFT,k\Delta t}\rangle$,

$$U_{eff,k\Delta t}|\Phi_{k-1}\rangle = \alpha_k|\Phi_k\rangle + \sqrt{1 - |\alpha_k|^2} |\phi_{correction,k\Delta t}\rangle, \quad (16)$$

and unitarity of $U_{eff,k\Delta t}$ preserves the norm of the second piece, $||U_{eff,k\Delta t}|\delta_{k-1}\rangle|| = \epsilon_{k-1}$. So

$$|\Psi_k\rangle - |\Phi_k\rangle = (\alpha_k - 1)|\Phi_k\rangle + \sqrt{1 - |\alpha_k|^2} |\phi_{correction,k\Delta t}\rangle + U_{eff,k\Delta t}|\delta_{k-1}\rangle, \quad (17)$$

and, by the triangle inequality,

$$\epsilon_k \leq |\alpha_k - 1| + \sqrt{1 - |\alpha_k|^2} + \epsilon_{k-1}. \quad (18)$$

With $|\alpha_k|^2 = 1 - e^{-S_k}$ (taking α_k real and positive without loss of generality), $\sqrt{1 - |\alpha_k|^2} = e^{-S_k/2}$ and $|\alpha_k - 1| = 1 - \sqrt{1 - e^{-S_k}} \approx \frac{1}{2}e^{-S_k} \ll e^{-S_k/2}$ for $S_k \gg 1$, so to leading order

$$\epsilon_k \lesssim \epsilon_{k-1} + e^{-S_k/2}. \quad (19)$$

Since $\epsilon_{-1} = 0$, induction gives

$$\epsilon_T \lesssim \sum_{k=0}^T e^{-S_k/2} \leq (T+1) e^{-S_{\min}/2}, \quad S_{\min} \equiv \min_{0 \leq k \leq T} S_k, \quad (20)$$

i.e.

$$\|U_{eff,T} \cdots U_{eff,0} |\psi_{EFT,0}\rangle - (U_{EFT})^{T+1} |\psi_{EFT,0}\rangle\| \lesssim (T+1) e^{-S_{\min}/2}, \quad (21)$$

which is $\ll 1$ precisely for $T \ll e^{S_{\min}/2}$, recovering the claimed approximate equality.

3 (2) The Born rule from maximally chaotic interactions

3.1 Interpreting Equation (2)

A maximally chaotic interaction Hamiltonian H_I , satisfying $[\hat{O}_S, H_I] = 0$, drives the joint system-environment state toward a Page-typical, near-maximally-entangled form [24, 25, 26]: an equal-amplitude superposition over N_E fine-grained branches, $|\Psi_f\rangle \approx \frac{1}{\sqrt{N_E}} \sum_k e^{i\varphi_k} |\psi_{S,k}\rangle |\phi_{E,k}\rangle$. The commutation condition $[\hat{O}_S, H_I] = 0$ guarantees that $\hat{O}_S$ -eigenvalue content is preserved along the evolution, so unitarity fixes the fraction of these N_E branches compatible with a given outcome λ_j to equal the initial coarse-grained probability $p_j = \langle \Psi_i | \hat{P}_S(\lambda_j) | \Psi_i \rangle$.

Because the N_E branches carry equal amplitude by construction, Zurek's envariance argument [27, 28] applies directly: each branch is equally likely. Summing over the $\approx N_E p_j$ branches compatible with outcome λ_j then gives the Born rule as a counting statement,

$$P(\lambda_j) = \frac{N_E p_j}{N_E} = p_j = |\langle \lambda_j | \Psi_i \rangle|^2, \quad (22)$$

rather than as an independent postulate. The Born rule has historically been introduced either as a bare postulate [29] or derived from noncontextuality assumptions on measurement outcomes [30]; the derivation here instead grounds it in the dynamics of a specific physical process. The improvement over Zurek's own construction is where the equal-amplitude decomposition comes from: Zurek fine-grains the state by hand, a step criticized as unmotivated, along with other aspects of the envariance argument, from several distinct directions [31, 32, 33, 34]. Here the same decomposition is instead a dynamical consequence of maximally chaotic interaction driving the state toward its Page-typical form - the fine-graining is dynamically derived, not assumed.

3.2 Ensemble theory in quantum black hole physics potentially hinting why the measurement basis gets picked as the pointer basis

Recently, there has been a resurgence of interest in how semiclassical gravity is dual to an ensemble of theories, averaged over. This view is sometimes invoked to argue that the reason the

semiclassical theory produces ever-increasing entanglement between the black hole interior and the exterior is that it is an ensemble theory. Once one passes to the actual, non-perturbatively corrected theory, final purity is restored for the black hole exterior.

In the measurement-basis context, this opens up the possibility that coarse-grained observables cannot distinguish these different theories, so that we are condemned to an ensemble view. In such a case, if the fine-grained pointer basis indexes theory randomness, we obtain a complete justification both for why the measurement basis is the pointer basis and for why we arrive at the final pure state.

In the double-scaled limit, this duality has been made precise for Jackiw-Teitelboim (JT) gravity. The disk-level JT path integral reduces to a Schwarzian boundary theory with density of states $\rho_0(E) \propto \sinh(2\pi\sqrt{E})$. Stanford and collaborators showed that this same density of states arises as the leading eigenvalue density of a Hermitian matrix integral $Z = \int dH e^{-L \text{Tr} V(H)}$, with $V(H)$ tuned so that its planar resolvent reproduces $\rho_0(E)$ exactly, and that JT's full sum over higher-genus topologies, including wormholes connecting multiple asymptotic boundaries, is reproduced order by order in $1/L$ by the matrix model's own genus expansion [1]. Connected multi-boundary correlators, which are exactly the ensemble variances of the matrix distribution, are computed on the gravity side by these wormhole geometries, and it is precisely these connected contributions that restore a unitary Page curve and produce the linear ramp of the spectral form factor [2]. Both consequences noted above, purity restored for the black hole exterior and coarse observables unable to distinguish the underlying ensemble, follow from this same matrix-model structure.

This part remains incomplete, with details from quantum gravity requiring completion. We leave these to future work.

4 Conclusion

The two results discussed in this paper arise from taking unitarity of an effective quantum theory seriously as a nontrivial constraint, which is almost always implicitly assumed but stronger than unitarity of the full quantum gravity theory.

For Equation (1), effective unitarity with the no-drama equivalence principle forces exponentially small deviations from semiclassical EFT for as long as both hold - so producing an $O(1)$ change in effective-theory entanglement entropy, as unitary black hole evaporation requires,

means giving up one of the equivalence principle, effective bulk Hilbert space factorization, or effective-side unitarity itself.

None of state-dependence, computational complexity, or the eigenstate thermalization hypothesis changes this. Our preferred route gives up the unmodified and (due to conservation of mass-energy) untenable equivalence principle only on a growing minority of microstates at any given time, leaving every individual branch, and hence locality, untouched.

For Equation (2), unitarity, applied to maximally chaotic interactions, turns the Born rule from a postulate into a counting consequence, closing the arbitrary fine-graining step in Zurek's envariance derivation - though it neither justifies the assumption of maximal chaos in a real measurement process nor resolves why any single outcome, rather than the full superposition, is realized. Effective unitarity, in both settings, does more work than it is usually given credit for.

Data availability and declaration of interests

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A Appendix: A sinner-priest dialogue summary of (1) - a layman version

A.1 Thesis

Effective-side unitarity plus a no-drama equivalence principle reproduces the standard semiclassical (Hawking) picture of black hole evaporation. Reversing that picture - so that the black hole exterior eventually purifies - therefore forces giving up at least one of three things: (A) Hilbert space factorization between the black hole interior and exterior, (B) the equivalence principle at the horizon, or (C) unitarity of the effective theory. The *superposition of smooth horizons* resolution shows (B) need not be rejected wholesale.

What follows is a story anyone with elementary geometry can follow. A sinner comes to confess his understanding of the universe to a priest.

A.2 Tale 1

Sinner: Here is what I believe, Father, and I want you to correct me. God (‘quantum gravity’) made the universe so that its states are rays of length one in His coordinate system (‘the quantum gravity Hilbert space’). He also gave us humans our own coordinate system (‘the effective bulk Hilbert space’), in which our states are likewise rays of length one. *The universe evolves by rotating these rays* - that is what dynamics means here.

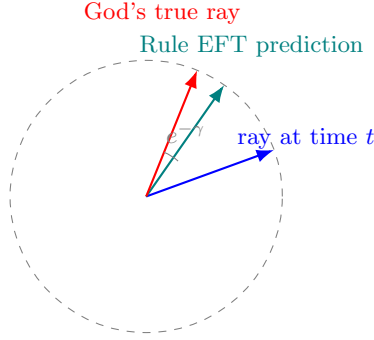


Figure 1: A ray of length one (blue) at time t rotates by time $t + \Delta t$. Rule EFT predicts the teal ray; God's true rotation (red) departs from it by only a tiny angle $e^{-\gamma}$ - the equivalence principle.

We have no access to God's own rule for rotating His rays ('the true quantum gravity theory') and are stuck working within our own coordinate system.

We did work out an approximate rotation rule for our effective rays - call it Rule EFT ('effective field theory'). It correctly rotates 'EFT-like' rays to good approximation, and we expect God's true rule to differ from it only by a tiny rotation of size $e^{-\gamma}$ ('equivalence principle,' since we cannot see the full truth behind His rule).

Over time these tiny corrections accumulate, so our ray gradually drifts from being purely EFT-like. But each correction is itself only a rotation - it cannot change a ray's length - so the accumulated drift stays small unless we wait an exponentially long time. We concluded Rule EFT should stay reliable for any practical purpose.

But some rays - the ones describing a black hole that has fully evaporated - must eventually have no relation between two subsystems (black hole interior and exterior), and Rule EFT forbids this outcome outright. Here I should be precise about what 'interior' means: our coordinate system is really built from *two* compartments joined together, one for the region outside the horizon ('exterior') and one for the region behind it ('interior').

Recently I heard sermons claiming we have misread God's word. The priests say (something like the eigenstate thermalization hypothesis) that God's rays look so different from our effective rays that, from His point of view, black hole rays either eventually reach zero interior coordinate or the interior and the exterior come to have no relation - full purification, no contradiction. Yet the priests also insist the equivalence principle stays holy: rays that were well-approximated by Rule EFT remain well-approximated by it. Working the rotations out myself, I cannot make these two claims fit together. But it is not my place to question God's messengers, so I have come to be corrected.

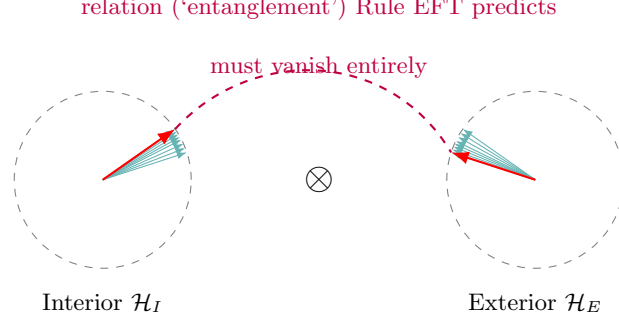


Figure 2: Small corrections (teal fans) only lightly rotate the interior and exterior rays each step, so the dashed link between them - the relation (‘entanglement’) between the two subsystems that Rule EFT predicts - survives almost unchanged, drifting only as much as the rays themselves do. Full purification requires this link to vanish entirely, not merely rotate: a qualitatively different, large change that many small corrections, bounded as in the text, cannot produce except over an exponentially long timescale.

Priest A: It is simple. You predict a nonzero interior coordinate or a relation that should not exist only because your observables are too coarse - Rule EFT misrepresents the relation (‘entanglement’) between the two subsystems in this regime.

Sinner: But God’s rule for converting His ray into ours is fixed - the coarseness of our observables should not matter to the ray itself. While humans may misrepresent (and falsely create) the (non-existent) relation, God knows what the true (non-)relation is in the human coordinate system. In other words, since God already knows that there is no relation (‘entanglement’) between two subsystems (‘interior and exterior’), He would not falsely create the relation in the human coordinate system - information constraint.

And the equivalence principle already guarantees that EFT-like rays stay well-approximated by Rule EFT within our own coordinate system. I even allowed God’s corrections to depend on the ray and on time. I still cannot see how the outcome can differ from Rule EFT’s prediction.

Priest B: There is a sudden tipping point in God’s rule, seen from our coordinate system: a discontinuous jump in the effective ray. The equivalence principle holds both before and after this jump. We understand it through error-correcting codes - right at the tipping point, the correction itself changes discontinuously. Call it a phase transition, if you like.

Sinner: But quantum mechanics - the rotation framework for both the human coordinate system and His coordinate system - forbids discontinuities in a finite system’s state. (This is supposedly only possible under quantum measurements.)

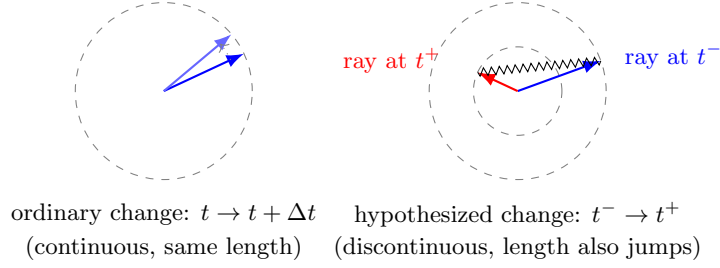


Figure 3: Ordinary rotation (left) moves a ray smoothly, along a continuous path, at fixed length. Priest B’s tipping point (right) instead proposes a jump - both direction *and* length change - with no continuous path of rays connecting t^- to t^+ (jagged connector). No such jump has ever been observed; ordinary quantum mechanics forbids it for a finite system.

And even setting that aside: linearity says that if a ray is a sum of two rays, its rotation is the sum of the two rotated rays, each keeping its own length. So is microscopic non-linearity now permitted, and is linearity no longer holy?

Priest C: Closer to the truth: your coordinate system is not actually fixed - its dimension can change over time. Usually it stays stable, but not always, and this alone is not yet the answer. For black hole rays specifically, the join of the ‘interior’ and ‘exterior’ compartments breaks down entirely: past some point there simply is no separate interior compartment left to have a coordinate in, significantly before the complete evaporation.

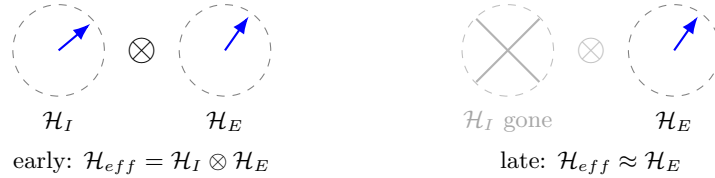


Figure 4: Priest C: Early on, interior and exterior are genuinely separate two-dimensional spaces $\mathcal{H}_I$ and $\mathcal{H}_E$, joined by $\otimes$; a ray has a well-defined interior coordinate because $\mathcal{H}_I$ exists to have a coordinate in. Late in evaporation $\mathcal{H}_I$ itself stops existing - factorization breaks down - so “interior coordinate” no longer makes sense. The size (dimension) of the interior and the exterior sub-space can vary over time.

Sinner: But Fathers (‘priests’) always perform what is often called the Page curve ritual - the relation (‘entanglement’) between the interior and the exterior increasing and then decreasing back to zero at the complete evaporation. What would that curve even mean, on your account? Was there never truly an interior coordinate to begin with?

Priest D: The equivalence principle is actually false at late times - it holds only as a property of an ensemble, not of any single actual ray. God permits us only probabilistic predictions about

our own effective ray, and for those predictions the equivalence principle holds on average. But for the one actual effective ray we live in, it may simply fail.

Heretic Priest E: There is no smooth horizon for the black hole interior in the first place and black holes are actually ‘fuzzballs’. The equivalence principle was false all the way from the beginning.

Partially heretic Priest F: Indeed, as you sinner points out, God admits no black hole interior even in our human coordinate system from the beginning. We are only reconstructing the black hole interior because we used to have general relativity. The reconstruction can be done, but we should not assume that rules of quantum mechanics are valid in the reconstruction. (‘Suvrat Raju’s holography of information’)

Sinner, worried that he may get excommunicated, addressing Priest D: And yet we have never actually witnessed the equivalence principle break down, even for black hole rays. I think the better path is not to reject it wholesale, but to refine exactly how much of it breaks, and where. Most of the ray keeps rotating under Rule EFT exactly as expected; only a shrinking sliver of it peels off entirely, into a ray with no interior left to speak of - a *superposition of smooth horizons*, rather than one horizon slowly and continuous evaporating. Each individual horizon branch still obeys its own equivalence principle. I came here to repent, so I should cut the details for now.

Sinner: And Priest D’s teaching, that probability enters only at the ensemble level, seems to me also the seed of why quantum mechanics gives us probabilities in the first place, rather than certainties.

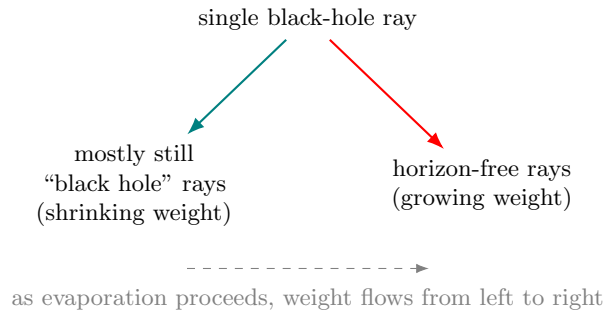


Figure 5: The escape route. A black-hole ray does not rotate all at once into a horizon-free ray; instead it splits, branch by branch. Most branches remain well-approximated by Rule EFT throughout (Figure 2); a growing minority peel off entirely into ordinary, horizon-free rays. Each branch, taken alone, keeps its own equivalence principle.

Moral. The whole disagreement in this tale turns on one fact: unitarity of the *human* coordinate system - ours, not God's - is a far stronger assumption than it looks, and quietly giving up factorization (Priest C), giving up the equivalence principle (Priest D - replaced with the equivalence principle as an average ensemble property), or giving up effective unitarity outright (Priest B - though only once temporarily) are the only three ways around it.

The tale's own preferred way out is the cheapest of the three: almost every branch keeps behaving exactly as Rule EFT expects, and only a growing few give up their horizon entirely, though details are cut in this tale.

A.3 Tale 2

The sinner meets the same priests again, questioned in a more mechanical language, and the tale ends at the same place Tale 1 did.

Sinner: Father, let me try to build the correction myself, so I can see where its smallness comes from.

Draw, in a two-dimensional coordinate system, a long ray from the origin, of length close to 1, and a second, much shorter ray, also from the origin, nearly perpendicular to the first. Call the long one L and the short one S ; think of L as the 'EFT-like' part of the state and S as everything Rule EFT leaves out. I want to fix a rule for how L and S rotate over time - the *Main Rule* - subject only to:

- (C1) At every moment, adding L and S together as a single ray gives something of length exactly 1.
- (C2) At every moment, the angle between L and S stays close to a right angle.
- (C3) At each step, L first rotates under some rule of its own - call it *Rule EFT*, since it plays exactly the role Rule EFT played in Tale 1 - which by itself would preserve L 's length. Then a small fraction a of L 's length is shaved off and reinvested as a new addition to S , orthogonal to the newly rotated L . The value of a is small but nonzero; otherwise it is free.
- (C4) Apart from (C1)-(C3), both Rule EFT's rotation of L and the exact direction in which S absorbs the shaved-off length are entirely free - fixed by God's rule, not by us.

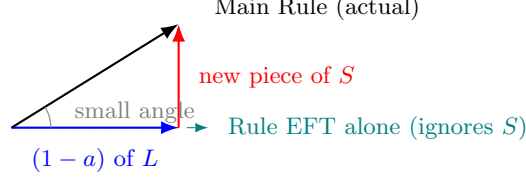


Figure 6: One step of the Main Rule (C3), drawn with a exaggerated for visibility. Rule EFT alone would send L straight to the dashed teal ray. The Main Rule instead shaves a fraction a off L 's length (blue) and reinvests it as a new orthogonal piece of S (red); the actual next ray (solid black) is their vector sum, still of length 1, sitting at a small angle from Rule EFT's prediction - the same $e^{-\gamma}$ -sized gap as Figure 1.

Given only (C1)-(C4) - not knowing Rule EFT, not knowing God's rule - how different can two things be: the ray you get by running the Main Rule (rotating L and S together, step by step, and adding them), versus the ray you get by running Rule EFT alone on L and ignoring S entirely? If a is as small as $e^{-(\text{something large})}$, the two stay close for an unimaginably long time; this follows from (C1)-(C4) alone, before anyone has said a word about what the Main Rule or God's rule actually is.

Priest A: And yet from God's standpoint, L and S are not really orthogonal at all - only *apparently* so, because human measurement is too coarse to see otherwise. Correct (C2), and the gap between the Main Rule's ray and Rule EFT's ray can open on an ordinary timescale, even while a stays small and L 's own length stays close to 1.

Priest A and Priest B: God gave humans a rule that predicts correctly almost all of the time, but at some moment there is a further rule only He can see, and the correct human-observable rays can only be derived from God's rays.

Sinner: But if a is small and L 's length stays close to 1, there is almost no room for S to have grown at all - so there is almost nothing for the angle between L and S to be wrong *about*. Drop (C2) entirely, and the Main Rule's ray still stays close to Rule EFT's prediction, for the same reason: (C3) is the only place growth enters, and (C3) alone already bounds it.

Furthermore, the Main Rule was never fully specified to begin with - (C4) largely leaves corrections free to depend on the ray and on time however God likes. A 'further rule only God can see' beyond that amounts to saying even God cannot produce the rotation humans actually observe, which cannot be right, since the Main Rule already has room for anything God's rule could throw at it without violating rotation-only dynamics ('unitarity of quantum mechanics') and the equivalence principle (Rule EFT).

Unless the lengths of L and S are significantly perturbed over time, which violates rotation-only rules and thereby quantum mechanics, there seems to be no way out.

Priest B: Then let me be specific: at some instant, God abruptly changes the *lengths* of L and S , as humans see them - though from God's own standpoint neither length changes, and His rule never changes either. Now some Main Rule predictions can differ significantly from the Rule EFT predictions.

(This is exactly the length-jump pictured in Figure 3, now applied to L and S specifically.)

Sinner: We have never once observed such a jump, and there is no reason to believe in it. Worse: if L and S 's lengths change with no continuous path between before and after, that is a result with no process behind it.

However smooth things look from God's side, if humans insist on a result with some process behind it ('causality'), then we are still condemned to Rule EFT, and we are forced to falsely conclude a relation between two subsystems (black hole interior and exterior, for example) that God Himself never sees.

Priest D: God does not give humans all the information there is. So humans cannot even predict what humans themselves will see - the Main Rule, as humans experience it, should be understood as a probability rule over which ray shows up. This shows why some Main Rule predictions can differ from what God actually predicts.

Sinner: Even so - can't God still predict exactly what humans will see externally? The bound I derived from (C1)-(C4) does not care whether the Main Rule is deterministic or probabilistic - it only used length and near-orthogonality. At this point, shouldn't we question how sacred Rule EFT is?

Priest B: Rule EFT is sacred, and so - at least almost - is the continuity of the human-visible ray. Priest D's suggestions are helpful but are not necessary.

My claim stands: God's world and the human world differ, so - apart from one temporary discontinuity - the human world stays continuous, and length stays conserved apart from that same discontinuity. That is how Rule EFT and the Main Rule are allowed to drift apart.

Unidentified Priest: We priests discuss these topics using the language of God ('quantum gravity'), not the language of sinners. I see no problem at all under the language of God. If you

want to discuss further, then you should provide derivations within God's language, or show why our best understanding of God's language is wrong. I am not willing to discuss further until you do so.